

# Message Queuing Telemetry Transport (MQTT)

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# MQTT Overview

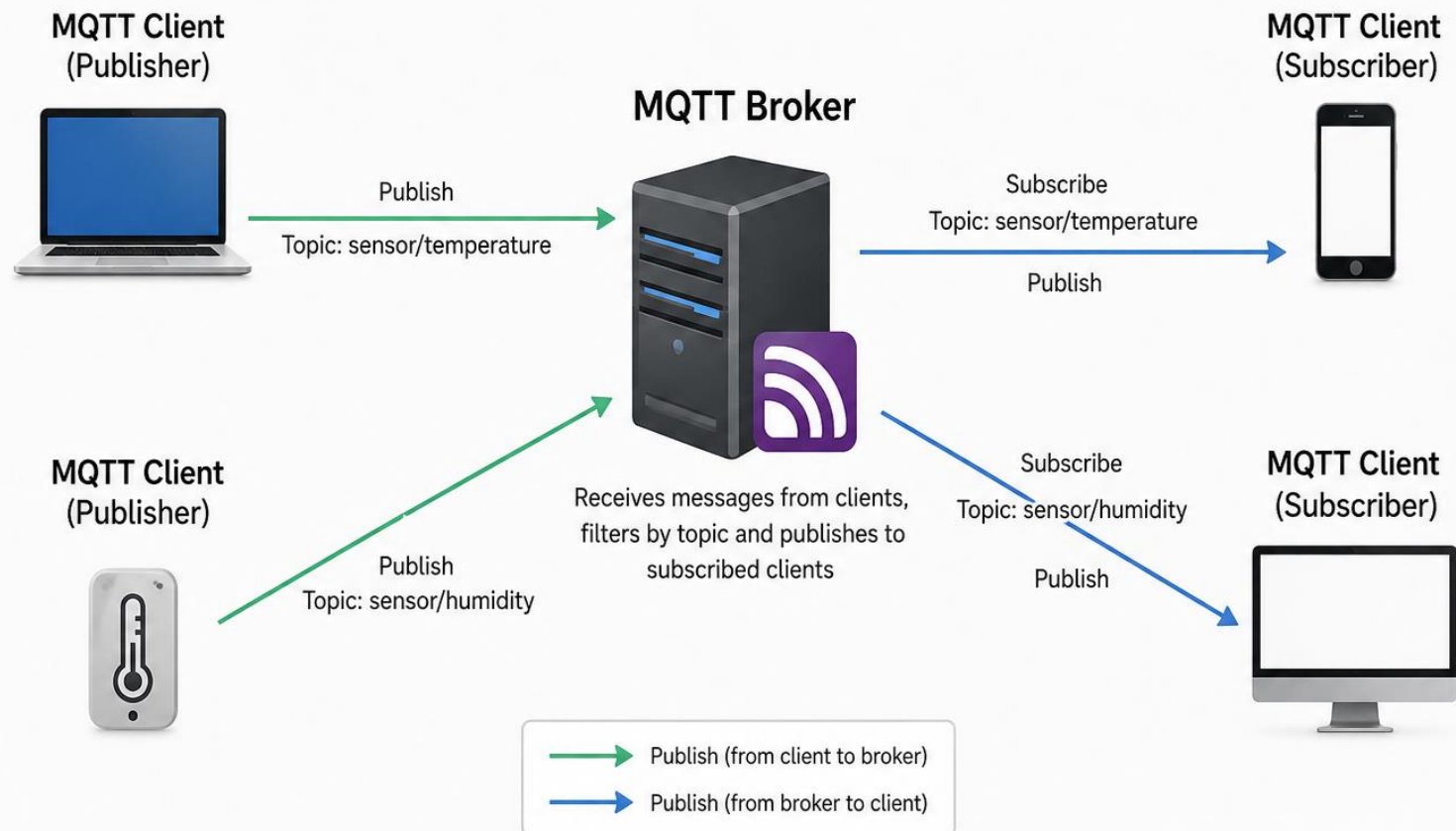
- MQTT, is a communications protocol **designed for Internet of Things** devices with extremely high latency and restricted low bandwidth.
- Message Queuing Telemetry Transport(MQTT) is a perfect protocol for **machine-to-machine (M2M) communication**.
- It is a TCP-based protocol relying on the publish-subscribe model.
- The publish-subscribe model involves multiple clients interacting with each other, without having any direct connection established between them.
- All clients communicate with other clients only via a third party known as a Broker.

# MQTT Client and Broker

- **Clients** publish messages on different topics **to brokers**. The **broker** is the **central server** that receives these messages and **filters** them based on their topics.
- It then sends these messages to **respective clients** that have subscribed to those different topics.
- The heart of any publish/subscribe protocol is the MQTT broker.
- Here the **broker is central hub** that receives messages, filters them, and distributes them to appropriate clients, such that both message publishers, as well as subscribers, are clients.

# MQTT Client and Broker

## MQTT Protocol – Client and Broker Model



# How MQTT Work

- MQTT's publish/subscribe (pub/sub) communication style, which aims to maximize available bandwidth, is an **alternative to conventional client-server architecture** that communicates directly with an endpoint.
- When a broker and a subscribing client lose contact, the broker will **store messages in a buffer** and send them to the subscriber whenever the subscriber is back up and running.
- MQTT clients, such as publishers and subscribers, can only speak with MQTT brokers. Any device or programme that runs a MQTT library can be a MQTT client, **ranging from microcontrollers like the Arduino to entire application servers housed in the cloud.**

# Features of MQTT

- **Lightweight:** MQTT is designed to be lightweight, making it suitable **for use in aid-restrained environments** inclusive of embedded systems and low-strength devices.
- **Publish-Subscribe Model:** In the publish-subscribe version, clients (publishers) send messages to subjects, and different clients (subscribers) acquire messages from subjects of interest.
- **Quality of Service (QoS) Levels:** MQTT supports exclusive stages of message **delivery warranty**.
- **Retained Messages:** MQTT lets in agents to store retained messages on topics, making sure that new subscribers acquire the maximum **latest message posted on a subject right now after subscribing**.
- **Security:** MQTT helps various protection mechanisms, consisting of Transport Layer Security (TLS) encryption and authentication mechanisms which include username/password and consumer certificates.

- Thank You